

**UNITED STATES OF AMERICA**  
**BEFORE THE FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads  
to the Interstate Transmission System

Docket No. RM26-4-000

**COMMENTS OF HERON POWER ELECTRONICS COMPANY**

**I. Introduction**

Heron Power Electronics Company ("Heron") respectfully submits these comments in response to the Commission's Notice Inviting Comments on the Secretary of Energy's Advance Notice of Proposed Rulemaking ("ANOPR") regarding the interconnection of large loads to the interstate transmission system.

Heron is a U.S. power electronics company developing solid-state transformer and grid-forming inverter technology well-suited to large load AI data center interconnections. Our primary product, the Heron Link, is a 5.5MVA medium voltage (34.5 kVAC) solid-state transformer that delivers 800 VDC output with integrated battery buffering to support grid ride-through, hold-up energy reserve, and resolve flicker concerns for datacenters. Heron has also developed OpenVSM, an open-source virtual synchronous machine software stack for grid-interactive inverters, enabling inverter-based resources and large power electronics loads to provide the same stabilizing grid services historically delivered by synchronous generators. The Heron Link will be manufactured domestically at scale, in a California facility producing 40GW of units annually.

Heron's comments focus on five themes that the Commission's current ANOPR framework does not fully address: (1) how the network risk of large datacenter loads can be mitigated by better power delivery architectures; (2) why energy-integrated power conversion, not only co-located generation, should be recognized as a valid compliance pathway for large load reliability requirements; (3) that grid-forming large loads differ fundamentally from passive loads and should be treated differently in interconnection and reliability rules; (4) how open, verifiable inverter control architectures reduce interconnection study burdens and build the operator trust that the interconnection queue currently lacks; and (5) how the Commission's technical standards directly shape the domestic power electronics manufacturing base that the Department of Energy is seeking to develop.

The stakes in this proceeding extend beyond interconnection procedure. Overly prescriptive rules with generation mandates or lengthy interconnection studies that fail to recognize the capabilities of modern power electronics to support the grid will produce longer queue timelines, higher costs, and more network upgrades. These outcomes would negatively impact ratepayers and slow the pace of AI infrastructure development that this administration has identified as a national priority. The Commission's rules in this proceeding will either advance that goal or constrain it. The technical pathway to interconnecting large AI loads quickly, affordably, and reliably exists today — but only if the Commission's framework is designed to accommodate it.

## **II. The Network Risk of Large Datacenter Loads Can be Mitigated by Better Power Delivery Architectures**

The Commission's ANOPR, NERC's Accelerated Large Load Action Plan, and NERC's Level 3 Alert issued May 4, 2026 converge on a common concern: AI data center loads introduce a category of grid risk not contemplated by existing interconnection standards.

First, datacenter loads don't ride through network disturbances. When grid voltage or frequency deviates beyond tight thresholds, data hall UPS systems immediately and completely disconnect compute load. In a grid with tens or hundreds of megawatts of co-located data center load, this behavior is self-reinforcing, and a minor grid disturbance can trigger simultaneous disconnection across multiple facilities, producing a sudden loss of load that exacerbates the very instability that triggered the disconnection. NERC's Level 3 Alert addresses this failure mode directly. Essential Action #1(e) further requires planners to model the voltage and frequency relay parameters that could isolate computational load from the bulk power system, reflecting NERC's recognition that the hair trigger protection settings embedded in existing data center electrical designs are themselves a reliability risk. The July 2024 Northern Virginia event, in which 1.5 GW of data center load disconnected simultaneously following a single upstream protection event, and multiple analogous incidents in ERCOT illustrate the system-wide consequences when datacenter UPS behavior at scale is left unmanaged.

Second, unlike traditional large loads, which draw power at relatively stable levels, unmitigated AI training workloads cause facility-wide load swings of 30 to 150 percent of rated power within milliseconds. Because all GPUs in a training cluster step through identical computational iterations in lockstep, these swings are synchronized across tens of megawatts simultaneously.

The Commission should understand, however, that these problems are not inherent to AI data centers. They are an artifact of legacy datacenter power delivery architectures. Conventional UPS systems instantaneously take datacenter load off-grid in the presence of minor voltage or frequency disturbances rather than ride-through, as power electronics-interconnected solar and batteries do. In the traditional 480 VAC power chain — medium-voltage transformer, low-voltage switchgear, UPS, and power distribution unit — AI training load transients pass through each conversion stage and propagate to the point of interconnection point largely unattenuated.

A different architecture can eliminate these problems at the source. Heron's SuperBBU is a high C-rate battery that couples directly to the data hall 800 VDC distribution power bus without an additional conversion stage, enabling ride-through support and sub-millisecond response to individual GPU-level load changes. Facility-wide training load swings that would otherwise propagate as severe flicker or medium-voltage current oscillations are absorbed within the 800 VDC bus before they reach the network point of interconnection. The facility's power draw can be smooth and ramp-controlled by the Heron Link solid-state transformer regardless of the AI workload power dynamics, supporting the objectives embodied in NERC's Level 3 Alert Essential Action #1. Additionally, the architecture directly addresses customer-side ride-through improvements that NERC's Level 3 Alert Essential Action #5 calls for, through hardware that makes destabilizing grid interactions physically impossible rather than behavioral commitments or operational coordination.

The Commission's evolving reliability framework should distinguish between data centers designed with integrated ride-through and load-smoothing capability and those without. Facilities that can demonstrate, at the point of interconnection, that their power draw remains within defined voltage & frequency ride-through, ramp-rate and ripple thresholds should be recognized as meeting the AI training load stability requirements the Commission and NERC are developing, without additional prescriptive equipment mandates.

### **III. Energy-Integrated Power Electronics Should Be a Recognized Compliance Pathway**

The ANOPR's framework for hybrid and co-located configurations (Principle 3) contemplates generation assets co-located with large loads as the primary mechanism for fault ride-through and load stability compliance. This is a valuable framework, but it describes only one of several technically sound compliance architectures.

Energy-integrated power electronics achieve the same reliability outcomes through a different mechanism: energy storage coupled directly to the datacenter 800 V DC distribution system tethered via grid-forming controls to the medium-voltage interconnection. A Heron Link-equipped facility illustrates this: the integrated SuperBBU storage provides 30 seconds of full-load hold-up; IEEE 2800 and ERCOT NOGRR 282 ride-through requirements are satisfied natively by the Heron Link hardware capability and grid-forming controls, without a separate generation asset on site.

The Commission should update its reliability compliance framework to explicitly recognize integrated power electronics solutions as a valid pathway alongside co-located generation. The relevant test should be performance-based: does the facility meet specified ride-through duration, ramp-rate, fault current, and voltage stability requirements at the point of interconnection? How it achieves those outcomes — through co-located generation, integrated storage, solid-state controls, or some combination — should be left to the developer's judgment. A prescriptive equipment-

based framework would unfairly disadvantage solutions that achieve equivalent or superior reliability outcomes through advanced power electronics rather than conventional generation assets.

Performance-based standards also reduce costs. Co-located synchronous generation assets such as gas peakers, microturbines, or diesel generators carry significant capital cost, permitting complexity, fuel supply requirements, and ongoing maintenance obligations. Where energy-integrated power electronics can satisfy the same performance requirements at lower cost, ratepayers benefit. FERC should write rules that allow the market to find that outcome.

#### **IV. Grid-Forming Large Loads Are a Grid Asset, Not Just a Reduced Liability**

Most commentary in this docket frames large loads as a problem to be managed: they stress the grid, shift costs to ratepayers, and require network upgrades to accommodate. The Commission's framework correctly seeks to mitigate these effects through readiness requirements, curtailability standards, and cost assignment principles. But this framing is incomplete.

A large load equipped with grid-forming controls is not merely a less-harmful version of a passive load but can be a positive grid asset. At the point of interconnection, a grid-forming inverter-based load behaves like a well-damped voltage-forming source — establishing a stable voltage and frequency reference, providing inertial response, delivering harmonic damping, and generating fault currents consistent with established protection relay coordination. These are the exact services the grid is losing as synchronous machines are retired and replaced by grid-following inverter-based resources and rectifier loads. NERC's own reliability assessments have identified declining synchronous inertia and the proliferation of grid-following IBRs as among the most significant near-term risks to bulk power system reliability. A large load that actively replenishes these services is not a burden on the grid, but a partial solution to the problem the Commission and NERC are spending considerable resources trying to solve through other means.

NERC's Accelerated Large Load Action Plan targets the reliability risks of grid-following loads. The Commission should recognize the inverse: grid-forming loads actively contribute to the reliability services that NERC's plan seeks to preserve. Grid-forming control architectures capable of delivering these services at the load interconnection point have been demonstrated on the generation side at utility scale, and are now being implemented in load-side power electronics like the Heron Link. The Commission need not wait for further technology development; it need only create the regulatory framework to recognize and reward performance that the market is already producing.

Two specific reforms would capture this value.

First, the Commission should create a distinct interconnection treatment pathway for large loads that demonstrate grid-forming capability — including extended ride-through, voltage support,

inertial response, and harmonic damping — at the point of interconnection. Such loads impose lower study burdens, require fewer network upgrades, and contribute to system reliability. The expedited study pathway the Commission is considering for flexible and curtailable loads is a natural starting point; grid-forming loads should qualify for comparable treatment.

Second, the Commission should direct RTOs and transmission owners to credit the grid services provided by grid-forming large loads against any direct-assignment network upgrade costs. The cost-causation principle underlying ANOPR Principle 8 should apply symmetrically to benefits as well as costs. A facility whose grid-forming controls demonstrably eliminate the need for a synchronous condenser, flicker mitigation device, or static VAR compensator provides a quantifiable benefit to the transmission system. Existing ratepayers benefit from that avoided investment. The Commission's rules should ensure that benefit is recognized in the interconnection cost allocation process.

## **V. Open, Verifiable Software Controls Reduce Interconnection Study Friction**

A significant driver of interconnection study cost and delay is the trust gap between inverter software control models and in-field behavior. Proprietary, nonlinear inverter controls are shared with system operators only as black-box models — opaque representations that may diverge from deployed firmware, and that must be re-evaluated whenever firmware is updated, plant design evolves, or the surrounding power system changes. This forces system operators into a conservative approach: extensive, bespoke electromagnetic transient (EMT) studies that cannot be reused across projects or updated without additional custom work. The resulting bottleneck is a structural contributor to the interconnection queue backlog across multiple RTOs.

Heron's OpenVSM initiative directly addresses this problem. OpenVSM is an open-source grid-forming controller whose architecture, tuning parameters, and embedded firmware code are publicly available. The open-source library includes reference models in Simulink, validated PSCAD model compatible with standard interconnection study workflows, embedded C code ready for direct compilation into inverter firmware by any manufacturer, and comprehensive technical documentation. Any developer, utility engineer, grid planner, or independent researcher can inspect, replicate, and validate its behavior. The controller's tuning parameters — inertial constant ( $H$ ), damping coefficient ( $D$ ), stator impedance ( $R+jX$ ), and fault current capability — are standard power system quantities that grid planners already use in synchronous machine studies.

OpenVSM is aligned with IEEE 2800 and the UNIFI consortium's ongoing grid-forming performance requirements. This means that a facility implementing OpenVSM-based controls can be studied using standardized templates rather than custom EMT models, and that those templates remain valid across firmware versions because the control algorithm is publicly documented, third-party verifiable, and its parameters and variables are field observable.

The Commission should incorporate principles into its final rule that help capture these benefits. RTOs and transmission providers should be directed to develop streamlined interconnection study procedures for loads implementing open, publicly documented control architectures, reducing the study depth required for facilities whose control behavior is fully verifiable. The Commission should further signal that technologies demonstrated to be compliant with open, validated control frameworks such as Open VSM are eligible to proceed through a streamlined interconnection study process. The Commission should also explore preferential queue treatment or reduced study deposit requirements for such facilities, recognizing that the study burden those facilities impose on operators is materially lower.

The trust gap between inverter model and field behavior is not a technical inevitability. It is a consequence of regulatory frameworks that do not reward openness. The Commission has the opportunity to restructure those incentives.

## **VI. FERC's Technical Standards Shape the Domestic Power Electronics Manufacturing Base**

The technical requirements the Commission establishes in this proceeding will define, for years to come, what equipment is necessary to interconnect a large load, what performance levels must be demonstrated, and which compliance pathways are available. In doing so, those rules will directly determine which equipment categories are purchased, and where from. This is not incidental to the Commission's mandate; it is a foreseeable and significant consequence of the Commission's regulatory choices.

The Department of Energy has identified advanced power electronics — including solid-state transformers, high-voltage DC distribution systems, and grid-forming inverters — as a manufacturing priority for the United States. Heron is developing U.S.-based manufacturing capability for the Heron Link and associated systems. The viability of that investment depends significantly on whether FERC's rules create demand for advanced solid-state power conversion technology, or whether rules are written around legacy equipment categories that have established global supply chains centered overseas.

The Commission should write performance-based standards that specify what grid behaviors large loads must exhibit — ride-through duration and depth, ramp-rate limits, fault current characteristics, voltage stability contributions — without specifying which equipment categories must be deployed to achieve them. This approach rewards innovation; any technology that meets performance requirements qualifies, creating a competitive market that drives cost reduction and capability improvement over time. It avoids locking in legacy equipment, and it creates the stable, technology-neutral demand signal that domestic manufacturers need to justify the capital investment required to build factory capacity at scale.

The Commission does not need to favor any particular manufacturer or technology. It need only ensure that its rules do not inadvertently foreclose advanced alternatives by writing requirements around conventional equipment categories. Performance-based standards aligned with IEEE 2800 and NERC's emerging computational load reliability criteria will achieve that outcome and create the demand pull that the U.S. advanced power electronics supply chain requires.

## **VII. Conclusion**

AI data centers need not be the grid stability problem that NERC's Level 3 Alert describes. With an appropriate energy-integrated power electronics architecture — grid-forming controls, integrated DC-bus storage, solid-state protection, and open, verifiable software — a large AI campus can be designed to absorb its own load volatility, provide inertial response and voltage support to the grid, ride through disturbances without burdening the system, and do all of this in a package that can be manufactured domestically at scale.

The Commission has an opportunity to write rules that make America's AI infrastructure buildout a source of grid strength rather than grid stress. Heron Power's technology is designed for exactly that outcome, and we look forward to supporting the Commission's work in this proceeding.

Respectfully submitted,

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